

WHEAT GERMINATION Monitoring System

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Wheat cultivation stands as one of the major staples in many countries, requiring meticulous attention to both temperature and humidity for optimal yields. The temperature range most conducive to ideal wheat seed germination lies between 20°C and 25°C. Unfortunately, untimely rainfall following the seeding process can hinder germination and foster seedling blight. As the crop progresses into the heading and flowering stages, extreme temperature fluctuations and drought can pose serious threats to its health. Cloudy weather, combined with high humidity and low temperatures, provides the perfect conditions for rust attacks. Furthermore, during the ripening phase, a steady average temperature of about 14-15°C is required for the wheat plant. Ensuring the right temperature conditions during grain filling and development is a crucial determinant of yield.

With this in mind, the present project was crafted to support farmers and wheat producers in effectively managing temperature and humidity for optimum production. The accompanying prototype, depicted in Fig. 1, illustrates the author's initial design. Essential components for this project are itemised in the bill of materials, as detailed in Table.

Circuit and working

The circuit diagram for the monitoring system, aimed at optimising wheat germination, is depicted in Fig. 2. The circuit's core revolves

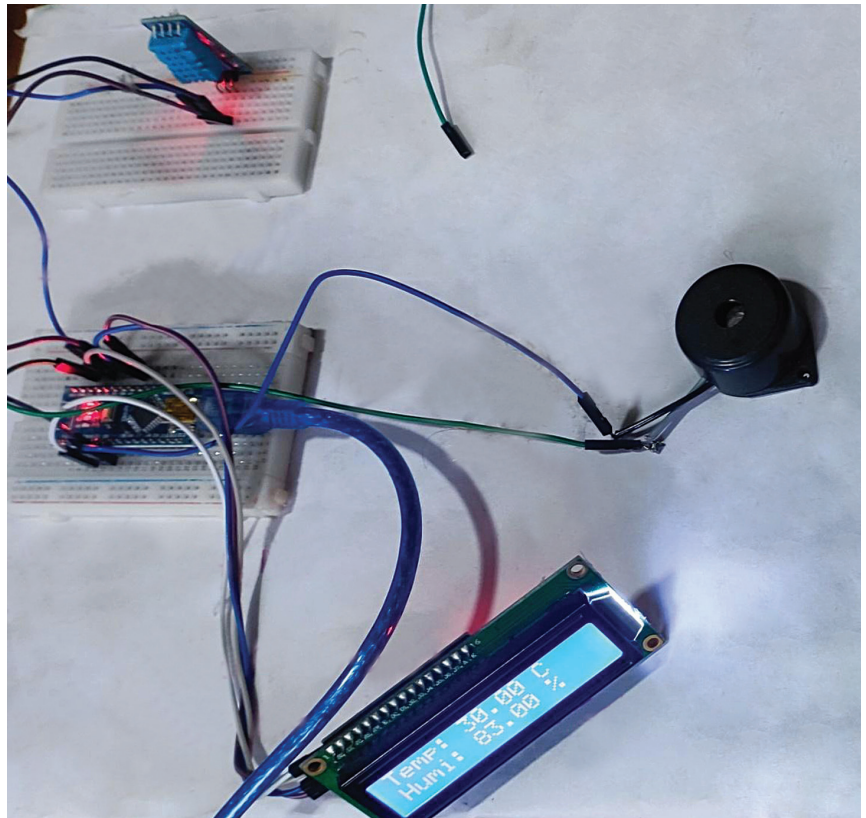


Fig. 1: Author's prototype

around the Arduino Nano (board1), an I2C LCD module (module2), a DHT11 humidity and temperature sensor (S1), jumper wires, a breadboard, a piezo buzzer, and additional essential components.

This circuit design draws heavily from Arduino Nano technology. A piezo buzzer has been integrated to sound an alarm when the temperature exceeds 25°C. The LCD screen is utilised to exhibit the field's temperature and humidity. The DHT11 sensor gauges the surrounding temperature and humidity, relaying the

BILL OF MATERIALS

Components	Quantity
LCD RG1602A I2C (MOD2)	1
Arduino NANO (MOD1)	1
Piezoelectric buzzer (P1)	1
DHT11 temperature sensor (S1)	1
5V power adaptor	1

data to the Arduino, which is then displayed on the connected LCD module. This system can be further enhanced by incorporating an ESP or other Wi-Fi modules to enable remote temperature monitoring.